

GitHub Pull Request (PR)

This document explains **what a Pull Request is**, **why it is used**, and **how to create it step by step** so that a student with **no prior knowledge** can follow and successfully generate a Pull Request.

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1. What is a Pull Request?

A **Pull Request (PR)** is a request to **merge changes from one branch into another branch** on GitHub.

- You make changes in a **separate branch**
- You ask GitHub to **pull your changes into the main branch**
- The project owner can **review, approve, or reject** the changes

Pull Request = Request to merge code

2. Why Pull Requests are Used

- To avoid directly changing the main branch
- To review code before merging
- To work safely in teams
- To track what changes were made

3. Basic Rule Before Creating a Pull Request

- **Never work directly on ***** main ***** branch**
- Always create a **new branch** for changes

METHOD 1: Creating Pull Request Using Git (Terminal + GitHub)

Step 1: Clone the Repository

```
git clone <repository-link>
cd repository-name
```

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Step 2: Check Current Branch

```
git branch
```

- * main → you are on main branch

Step 3: Create a New Branch

```
git checkout -b feature-update
```

- Creates a new branch
- Switches to that branch

Step 4: Make Changes in Files

Example:

- • README.
 md
Modify
OR add a new file index.c

Step 5: Check File Status

```
git status
```

Step 6: Add Changes to Staging Area

Add one file:

```
git add README.md
```

Add all files:

```
git add .
```

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Step 7: Commit Changes

```
git commit -m "Added feature update"
```

Step 8: Push Branch to GitHub

```
git push origin feature-update
```

Your branch is now available on GitHub

Step 9: Create Pull Request on GitHub

1. Open GitHub repository
2. You will see "**Compare & pull request**" button
3. Click it
4. 5.

branch:	main
Base	
- Compare branch: feature-update
6. Add title and description
7. Click **Create pull request**

Step 10: Pull Request Created

- Changes are waiting for review
- Can be merged after approval

METHOD 2: Creating Pull Request Using GitHub Website Only (No Git Commands)

Step 1: Create a New Branch on GitHub

1. Open repository
2. Click branch dropdown (usually shows main)
3. Type new branch name
4. Click **Create branch**

Step 2: Make Changes on GitHub

1. Open a file OR create a new file
2. 3. directly on
Edit file GitHub
- Scroll down
4. Select **Commit to new branch**
5. 6. commit
Enter message
- Commit changes

Step 3: Open Pull Request

1. GitHub will prompt to **Create Pull Request**
2. Click **Compare & pull request**
3. Verify base = main
4. Verify compare = new branch
5. Click **Create pull request**

4. Merging a Pull Request

Merge Options

- | | |
|-------------------|---|
| 1. 2. | request – normal merge |
| Merge pull | |
| | Squash and merge – combine commits |
| 3. | merge – clean |
| Rebase and | history |

For beginners: Use **Merge pull request**

Steps to Merge

1. Open Pull Request
2. Click **Merge pull request**
3. Click **Confirm merge**

5. After Merge (Cleanup)

Delete branch (optional):

- Click **Delete branch** button

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6. Pull Request Flow Summary

Create Branch → Make Changes → Commit → Push → Create PR → Review → Merge

7. Common Beginner Mistakes

- Working directly on main
-

Forgetting to push branch

-

Creating PR from main to main

8. Essential Pull Request Commands (Cheat Sheet)

git branch # list branches

git checkout -b branch-name # create & switch branch git add . # stage changes

git commit -m "message" # commit changes

git push origin branch-name # push branch to GitHub If these steps are followed in order, a

Pull Request can be created successfully without errors.